Pi-Lomar Mini Observatory																				
Source																		R1,409.70		
1	https://www.instructables.com/Pi-lomar-3D-Printed-Working-Miniature-Observatory-/																			
	Exchange Rate																			
									R1,409.70											

Item #	Description / Value	Category	Component	Quantity	Mandatory	Price Per Unit (ZAR)	Import Duties Per Unit (ZAR)	Delivery Cost Per Unit (ZAR)	Total Cost (R)	Aquired	Arrived	Recommended purchae link	Source	Available purchase link 1	Source	Available purchase link 2	Source	Available purchase link 3	Source
1	Raspberry Pi 4B. At least 2GB, preferably 4GB	Electronics	Motorcontroller Board	1	☑	R0.00	R0.00	R0.00	R0.00	☑	☑		☐	PiShop	☐		☐		☐
2	32GB micro SD card	Electronics	Motorcontroller Board	1	☑	R0.00	R0.00	R0.00	R0.00	☑	☑		☐	PiShop	☐		☐		☐
3	Raspberry Pi USB power supply	Electronics	Motorcontroller Board	1	☑	R0.00	R0.00	R0.00	R0.00	☑	☑		☐		☐		☐		☐
4	Pimoroni Tiny2035 microcontroller 8MB Headered version	Electronics	Motorcontroller Board	1	☑	R189.90	R0.00	R100.00	R289.90	☑	☑	Pimorony	☐	Pishop	☐		☐		☐
5	0.9 Degree Nema17 stepper motor (eg uk_stepperonline) 17HM19-2004S (x2). (1A motors work fine, but I use 2A motors so that there's more power available in the future)	Electronics	Motorcontroller Board	2	☑	R399.95	R0.00	R50.00	R899.90	☑	☑	Stepper Online UK	☐	3d Printing Store	☐		☐		☐
6	DRV8825 stepper motor driver board	Electronics	Motorcontroller Board	2	☑	R109.95	R0.00	R0.00	R219.90	☑	☑	Pololu	☑	3d Printing Store	☐	Pishop	☐		☐
7	47uF capacitors 25V	Electronics	Motorcontroller Board	3	☑	R1.29	R0.00	R0.00	R0.00	☑	☑		☐	DIY Electronics	☐	Communica	☐		☐
8	BC548CTA NPN transistor	Electronics	Motorcontroller Board	1	☑	R0.00	R0.00	R0.00	R0.00	☑	☑		☐	DIY Electronics	☐		☐		☐
9	40way ribbon cable	Electronics	Motorcontroller Board	1	☑	R0.00	R0.00	R0.00	R0.00	☑	☑		☐		☐		☐		☐
10	40way PCB header	Electronics	Motorcontroller Board	1	☑	R0.00	R0.00	R0.00	R0.00	☑	☑		☐		☐		☐		☐
11	Wire	Electronics	Motorcontroller Board	1	☑	R0.00	R0.00	R0.00	R0.00	☑	☑		☐		☐		☐		☐
12	Prototyping board	Electronics	Motorcontroller Board	1	☑	R0.00	R0.00	R0.00	R0.00	☑	☑		☐		☐		☐		☐
13	320 to 680 Ohm resistor	Electronics	Motorcontroller Board	1	☑	R0.00	R0.00	R0.00	R0.00	☑	☑		☐		☐		☐		☐
14	10K Ohm resistor	Electronics	Motorcontroller Board	1	☑	R0.00	R0.00	R0.00	R0.00	☑	☑		☐	DIY Electronics	☐		☐		☐
15	12V 8A power supply	Electronics	Motorcontroller Board	1	☑	R0.00	R0.00	R0.00	R0.00	☑	☑		☐	AC DC	☐		☐		☐
16	250mm Lazy Susan bearing.	Mechanical	Azimuth Drive And Foundation	1	☑	R0.00	R0.00	R0.00	R0.00	☑	☑		☐	Perfect Tech SA	☐		☐		☐
17	Steel Rod Diameter 5mm Diamter 300mm Length	Mechanical	Azimuth Drive And Foundation	1	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐	Ali Express	☐		☐		☐
18	Steel Rod Diameter M5 Diamter 22mm Length	Mechanical	Azimuth Drive And Foundation	6	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐	Ali Express	☐		☐		☐
18	60 Teeth Worm Gear Bore 5mm	Mechanical	Azimuth Drive And Foundation	2	☑	R0.00	R0.00	R0.00	R0.00	☐	☐	Ebay	☐	Ali Express	☐		☐		☐
19	M4 countersunk screws	Mechanical	Azimuth Drive And Foundation	8	☑	R0.00	R0.00	R0.00	R0.00	☑	☑		☐		☐		☐		☐
19	M5 bolts 25mm	Mechanical	Azimuth Drive And Foundation	30	☑	R0.00	R0.00	R0.00	R0.00	☐	☐	Ebay	☐	Ali Express	☐		☐		☐
20	M5 bolts 30mm	Mechanical	Azimuth Drive And Foundation	30	☑	R0.00	R0.00	R0.00	R0.00	☐	☐	Ebay	☐	Ali Express	☐		☐		☐
20	M5 6x19mm skateboard/RC bearings	Mechanical	Azimuth Drive And Foundation	15	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
21	M6 6x19mm skateboard/RC bearings	Mechanical	Azimuth Drive And Foundation	5	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
19	M4 and M5 nuts and bolts 12-30mm	Mechanical	Azimuth Drive And Foundation	1	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
22	M6 flange coupler	Mechanical	Azimuth Drive And Foundation	1	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
21	M5 and M6 collars.	Mechanical	Azimuth Drive And Foundation	1	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
22	B.13.2.AzimuthDriveM5.X1.stl	3D Printed	1-azimuth-drive-wheel	1	☑	R0.00	R0.00	R0.00	R0.00	☐	☐	Instructables	☐		☐		☐		☐
20	B.13.2.AzimuthDriveM6.X1.stl	3D Printed	1-azimuth-drive-wheel	1	☑	R0.00	R0.00	R0.00	R0.00	☐	☐	Instructables	☐		☐		☐		☐
23	A.13.6.Foundation.X9.stl	3D Printed	2-foundation	9	☑	R0.00	R0.00	R0.00	R0.00	☐	☐	Instructables	☐		☐		☐		☐
22	AC.13.6.AdjustableFoundation.X3.stl	3D Printed	2-foundation	3	☑	R0.00	R0.00	R0.00	R0.00	☐	☐	Instructables	☐		☐		☐		☐
23	AG.13.6.RingGear.X1.stl	3D Printed	2-foundation	1	☑	R0.00	R0.00	R0.00	R0.00	☐	☐	Instructables	☐		☐		☐		☐
21	AE.13.7+FootRing.X1.stl	3D Printed	2-foundation	1	☑	R0.00	R0.00	R0.00	R0.00	☐	☐	Instructables	☐		☐		☐		☐
24	AB.13.2.AdjustableFoot.X3.stl	3D Printed	2-foundation	3	☑	R0.00	R0.00	R0.00	R0.00	☐	☐	Instructables	☐		☐		☐		☐
23	38mmPeg.X48.stl	3D Printed	2-foundation	48	☑	R0.00	R0.00	R0.00	R0.00	☐	☐	Instructables	☐		☐		☐		☐
24	E.13.6.DomeBearingBase.X1.stl	3D Printed	3-platform	1	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
25	D.13.5.DomeBase.X10.stl	3D Printed	3-platform	10	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
26	DaM4.X10.stl	3D Printed	3-platform	10	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
27	DaM5.X10.stl	3D Printed	3-platform	10	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
28	G.13.1.WallMount.X2.stl	3D Printed	3-platform	2	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
29	38mmPeg.X30.stl	3D Printed	3-platform	30	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
30	18mmPeg.X30.stl	3D Printed	3-platform	30	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
31	P.14.0.InnerWall.X2.stl	3D Printed	4-camera-tower	2	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐

31	P.13.6.InnerWall.X2.stl	3D Printed	4-camera-tower	2	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
32	Y.13.1.CableRing.X1.stl	3D Printed	4-camera-tower	1	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
33	WL.13.2.LensCradle.X1.stl	3D Printed	4-camera-tower	1	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
34	W.13.1.CameraCradle.X1.stl	3D Printed	4-camera-tower	1	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
35	GA.13.1.Brace.X1.stl	3D Printed	4-camera-tower	1	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
36	R.13.3.AltitudeHub.X2.stl	3D Printed	4-camera-tower	2	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
37	L.13.4.AzimuthBrace.X2.stl	3D Printed	4-camera-tower	2	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
38	G.13.1.WallMount.X2.stl	3D Printed	4-camera-tower	2	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
39	JB.13.7.AzimuthMotor.X1.stl	3D Printed	5-azimuth-gearbox	1	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
40	T.13.8.AltitudeBracket.X1.stl	3D Printed	6-altitude-gearbox	1	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
41	JA.13.7.AltitudeMotor.X1.stl	3D Printed	6-altitude-gearbox	1	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
42	S.13.2.12TM1-5mm.X1.stl	3D Printed	6-altitude-gearbox	1	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
43	S.13.2.12TM1-6mm.X1.stl	3D Printed	6-altitude-gearbox	1	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
44	NA.14.3.MicrocontrollerHolder.X1.stl	3D Printed	-	1	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
45	Wall.13.2.X8.stl	3D Printed	7-dome	8	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
46	Dome.13.1.Segment.X8.stl	3D Printed	7-dome	8	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
46	Dome.13.3.CapAR.X1.stl	3D Printed	7-dome	1	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
47	Dome.13.3.CapAL.X1.stl	3D Printed	7-dome	1	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
51	Dome.13.1.SideA1.X1.stl	3D Printed	7-dome	1	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
50	Dome.13.1.SideA2.X1.stl	3D Printed	7-dome	1	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
49	Dome.13.1.SideB1.X1.stl	3D Printed	7-dome	1	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
48	Dome.13.1.SideB2.X1.stl	3D Printed	7-dome	1	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
52	13.2.DoorA1.X1.stl	3D Printed	7-dome	1	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
53	13.2.DoorA2.X1.stl	3D Printed	7-dome	1	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
54	13.2.DoorB1.X1.stl	3D Printed	7-dome	1	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
55	13.2.DoorB2.X1.stl	3D Printed	7-dome	1	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
56	78mmPeg.X24.stl	3D Printed	7-dome	24	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
57	38mmPeg.X8.stl	3D Printed	7-dome	8	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
58	18mmPeg.X36.stl	3D Printed	7-dome	36	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐
59		3D Printed	7-dome	1	☑	R0.00	R0.00	R0.00	R0.00	☐	☐		☐		☐		☐		☐